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# SOFTWARE ASSISTED DEVELOPMENT OF PROTECTION COORDINATION CONCEPTS IN NATIONWIDE POWER SYSTEMS

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## Abstract

This paper is dealing with an urgent task of protection coordination in nationwide power systems. The coordination of overcurrent protection is strongly depending on the network configuration in contrast to distance or differential protection. Proposed concepts should be evaluated applied to the network. Nowadays numerical simulation tools like PSS™SINCAL provide the possibility of a holistic simulation of network and protection behaviour.
Different overcurrent protection concepts for transmission networks are analysed on their feasibility. The software assistance and the visualisation of the enormous amount of output data are described. The evaluation shows clearly the difference and pros and cons between the discussed protection concepts. Generic network configurations leading to maloperation can be clearly found. Further on the possibility of the development of systematic solutions covering the identified practical constraints is provided.

## 1 Introduction

The complexity of power systems is increasing world wide because of permanent network extensions and ongoing interconnections. An important factor for blackout prevention is a regular review and if required an adjustment of the protection coordination concept. In the field of transmission networks, protection coordination is developing more and more towards a task of handling nationwide power systems under continuous change and with numerous protection devices preferably graded by software tools.
The determination of an appropriate protection coordination concept of overcurrent-time (O/C) protection for phase-to-earth (line residual O/C relays 67N) and phase-to-phase faults (O/C relays 51) is vital for the operation of transmission networks. Line residual O/C relays are necessary as back-up protection for remote earth faults and as protection against high resistance faults not detected by distance relays. Phase-to-phase O/C relays play the role of the final reserve for long time delayed back-up protection.

In meshed systems a general analytical solution for grading O/C relays, ensuring 100% selectivity and high speed fault clearing, cannot be found in all cases. The coordination has to be robust against changing network conditions including grid topology, short-circuit and infeed behaviour. That is why the software assisted development method must provide the possibility of the evaluation and comparison of different proposed solutions. The objectives of this paper can be given as follows:

- design and evaluation of a O/C protection coordination concept based on equal characteristics for all O/C relays applied to be as robust as possible
- different IEC/ANSI characteristics and a non-standardised logarithmic characteristic should be used
- create and use of a software solution and environment which is able to cover the above mentioned requisites
- evaluation and comparison of different results in one diagram assessing selectivity in the whole network
- identification of practical constraints in the application of the proposed protection coordination concept for generic network configurations.

## 2 O/C protection coordination concept

### 2.1 A proven protection coordination concept in distribution networks adopted to transmission networks

Low voltage distribution networks, especially in regions with high load density, are often strongly meshed and subject to unpredictable network configuration changes. General regular update of the O/C protection concepts seems not be worthwhile the effort. In praxis a rather simple but nevertheless effective concept based on absolutely identical high-performance fuses at each feeder is applied. Thereby selectivity is ensured only by means of a sufficient difference in current which results in a sufficient clearing time interval.
Figure 1 and Equation 1 show the underlying fundamentals in meshed networks: the short-circuit current seen by relay R1 on the faulty line is considerably the highest, since it is the total of all feeding short-circuit currents.

$$\sum_{i=1}^{n} k_i = 1 \qquad (1)$$

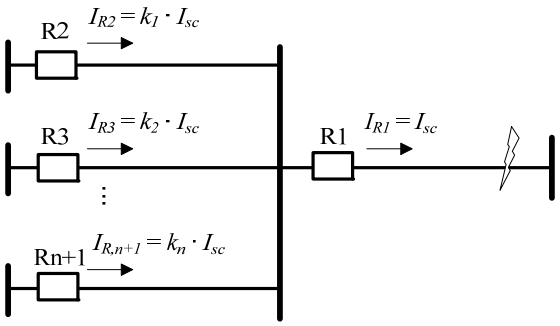



Figure 1: Short-circuit current in meshed networks

As the tripping time $t_p$ is a function of the short-circuit current $I_{sc}$ a desired clearing time interval (*CTI*) is kept if the applied characteristic satisfies the following inequation

$$t_p\left(k_{\max} \cdot I_{sc}\right) \geq t_p\left(I_{sc}\right) + CTI \qquad (2)$$

Whereas meshed distribution networks have to be considered as rather homogeneous for the fuse concept to work, nationwide transmission networks might pose novel aspects which have to be discussed and covered by the adopted O/C protection coordination concept:

A. a very high diversity in adjacent line lengths
B. different power ratings for different connections
C. single- and double-circuit line sections side by side
D. local bulk generation

Consequently the question arises if a similar concept, using O/C relays instead of fuses, can be applied to high voltage transmission networks?

### 2.2 Basic setting-rules of O/C characteristics

Following the described concept above, the identical characteristic must be applied to all O/C relays throughout the transmission network. The standardised ANSI and IEC characteristics are commonly used. To set their pickup current $I_p$ and their time multiplier $T_p$ two basic setting-rules are essential:

1. Because the range of inverse operation of ANSI/IEC characteristics is between $I_p$ and $20 \cdot I_p$, $T_p$ can be calculated at $20 \cdot I_p$, regarding a minimum tripping time $t_{p,min}$ to be given from the coordination with unit protection devices.
2. $I_p$ depends on the thermal limiting current $I_{th,min}$ of the weakest transmission line multiplied by a security factor *s* (Equation 3).

$$I_p = s \cdot I_{th,\min} \qquad (3)$$

With this approach lines with higher loadability $I_{th} > I_{th,min}$ need individual pickup values $I_{p,ind}$ considering the higher thermal limiting current (Equation 4).

$$I_{p,ind} = s \cdot I_{th} \qquad (4)$$

Consequently, the characteristic of these lines will be cut off from the left side to prevent false tripping in load cases (Figure 2a).

Reference [1] describes a non-standardised logarithmic characteristic. Its advantages are a (user-defined) wider range of inverse operation and a better performance regarding selectivity. Therefore it seems suitable for this protection coordination concept. The logarithmic characteristic is defined by the following Equation 5

$$t = a \cdot \log\left(\frac{I}{I_p}\right) + b \qquad (5)$$

The basic setting-rules for the parameter $I_p$ and $b$ can be comparably adopted considering the wider range of inverse operation (Figure 2b).

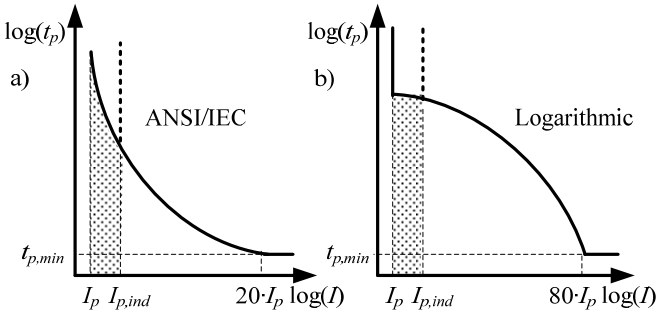



Figure 2: Exemplary O/C characteristics

The choice of slope $a$ such as the choice of concrete ANSI/IEC characteristics depend on the maximum tripping time to be allowed. It can be observed the steeper the gradient of the characteristic the higher the maximum tripping time and the better the performance regarding selectivity. This is discussed in the following chapter.

### 2.3 Suitability of different O/C characteristics

The suitability of different O/C characteristics can be assessed determining a selectivity-performance indicator defined below. As the identical characteristic is applied to all O/C relays, only accommodated to local conditions by the parameter $I_{p,ind}$, the characteristics of all relays are 100% congruent. Therefore general statements towards selectivity in respect of a required minimum current difference $I_{diff,min}$ can be made.

Be $t_p = f(I / I_p)$ and $I / I_p = f^{-1}(t_p)$ the tripping characteristic and its inverse function. The required $I_{diff,min}$ between relay R1 and upstream relay R2 (Figure 1), needed to ensure the clearing time interval *CTI* and considering a supposed current measurement tolerance *TOL*, can be mathematically described with Equation 6. It defines the selectivity-performance indicator from which can be seen that the lower $I_{diff,min}$ the better the performance regarding selectivity.

$$\frac{I_{diff,\min}}{I_{R1}} = \frac{I_{R1} - I_{R2}}{I_{R1}} = 1 - \frac{f^{-1}\left( f\left( (1-TOL)\frac{I_{R1}}{I_p} \right) + CTI \right)}{(1+TOL)\left( \frac{I_{R1}}{I_p} \right)} \quad (6)$$

Figure 3 and Figure 4 show the selectivity-performance indicator according to Equation 6 of the ANSI Short-Inverse characteristic and of the IEC Very-Inverse characteristic respectively. One can see that the selectivity gets better the steeper the characteristic against lower short-circuit currents. In comparison with both characteristics it is also observable that the relatively steep IEC Very-Inverse characteristic has an overall better performance regarding selectivity (i.e the required minimum current difference is low).
In contrast to standardised characteristics the logarithmic one features one peculiarity, which is on closer examination useful especially for this protection coordination concept: the resultant required minimum current difference is a constant value, thus independent of the short-circuit current. It can be mathematically described as

$$\frac{I_{diff,\min}}{I_{R1}} = 1 - k_{\max} = 1 - \frac{1-TOL}{1+TOL} \; e^{\ln(10)\frac{CTI}{a}} \quad (7)$$

and directly determines the coefficient $k_{max}$ as defined in chapter 2.1 (Figure 3b).

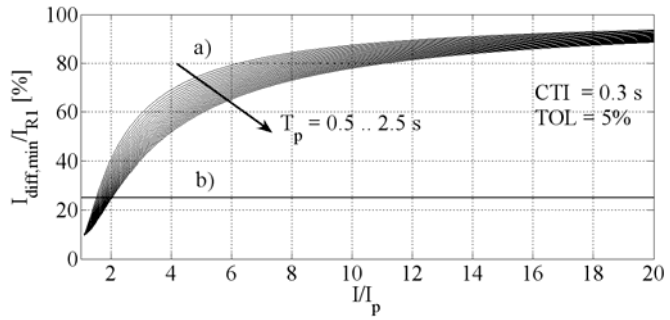



Figure 3: Selectivity-performance indicator of the
a) ANSI Short-Invers characteristic;
b) logarithmic characteristic ($a$ = -3.61)

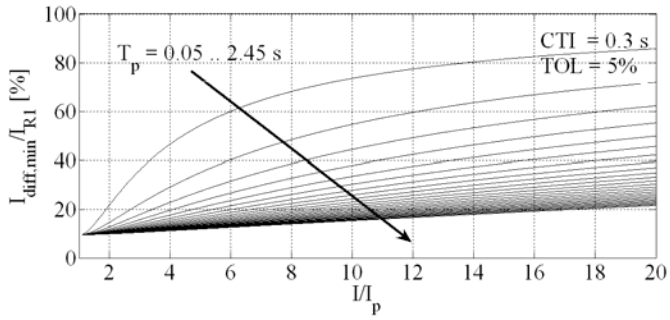



Figure 4: Selectivity-performance indicator of the IEC Very-Invers characteristic

In general, coordination of protection relays with an identical characteristic requires a minimum difference in measured short-circuit current. Based on the experiences obtained, some preliminary conclusions can be drawn:

- The gradient of the examined characteristic has an all-dominant influence on selectivity. Standardized ANSI/IEC-characteristics are getting significantly better towards low short-circuit currents, whereas logarithmic characteristics stay unaffected hereof.
- The higher the time multiplier the lower the required difference in current and the higher the maximum tripping time.
- The pickup value is also very important for protection relay coordination. A higher one leads to an increased inverse operation range and shifts the operating point towards the favourable steeper area of the characteristic.

## 4 Software Solution

In order to assess the applicability and performance of the proposed protection coordination concept in real transmission networks a simulation environment was set up. Simulations were carried out using the standard mathematic software MATLAB® in combination with the practice-proven power system planning and simulation software PSS™SINCAL [2]. The latter one was used for power flow, short-circuit and stepped-event protection simulation. PSS™SINCAL's protection device database includes all commonly used protection characteristics and relays from various manufacturers. Data is stored in an open MS Access database and is completely editable via a standard Open Database Connectivity (ODBC) interface. This enabled efficient batch-processing of the large-scale computationally intensive protection simulation.

### 4.1 Program sequence

For protection coordination examinations of an existing network model, faults have to be placed on every transmission line by hand. Detailed fault conditions can then be manipulated via ODBC by MATLAB® as described above. Knowing the exact assignment of lines and protection relays as stored in the database, a loop algorithm sweeps the fault location from one end to the other of all transmission lines. With every loop PSS™SINCAL is automatically called via Windows Scripting Host to do the stepwise fault clearing calculations. Via ODBC the results come back to MATLAB® to be further processed. In this way detailed information about selective fault clearing for a fault at every single point of the network can be generated and visualised.

### 4.2 Visualisation of simulation results

The enormous amount of output data generated has to be illustrated in a way that allows an easy evaluation of results. For that reason the following illustration method was chosen.
The different transmission lines are plotted on the y-axis whereas the x-axis represents the fault location related to line length. The colour value at every single point expresses the (selective) fault clearing time for a fault at this location. Moreover there are two special conditions: a white bar with

two lines indicates non-cleared faults and a black-white bar represents unselective fault clearing. In this way protection coordination of widespread networks can be visually captured and assessed in one diagram (Figures 5-8).

## 5 Evaluation of the proposed protection coordination concept

In this way the development tool was used to evaluate the proposed protection coordination concept. A real transmission network was chosen and implemented in PSS™SINCAL which considers the aspects A. to D. in chapter 2.1 to achieve practical statements.

It is fed by three infeeds (short-circuit power $S_{sc1}$"= 1000 MVA, $S_{sc2}$"= 5000 MVA and $S_{sc3}$"= 133 MVA) representing adjacent transmission networks and by four power plants (rated power $S_{rG1}$ = 4x65 MVA, $S_{rG2}$ = 4x210MVA, $S_{rG3}$ = 2x81 MVA and $S_{rG4}$ = 2x81 MVA). Except of three radial feeders, the network is meshed and consists of 23 transmission routes. Twenty routes are double-circuit lines and three are single-circuit lines. Four line types are applied (Table 1). The network is solidly earthed.

| No. | Name | $R$'[Ω/km] | $X$'[Ω/km] | $I_{th}$ [kA] |
|---|---|---|---|---|
| 1. | GOOSE | 0.1043 | 0.4231 | 0.580 |
| 2. | COOT | 0.0430 | 0.3108 | 1.250 |
| 3. | REDWING | 0.0448 | 0.3031 | 1.250 |
| 4. | XLPE Cable | 0.0287 | 0.1830 | 0.884 |

Table 1: Applied line types

### 5.1 Phase-to-phase faults

For phase-to-phase faults O/C relays play the role of the final reserve. They are typically non-directional (51). From the suitability performance assessment (chapter 2.3) the IEC Very-Inverse characteristic and the logarithmic characteristic stand out and are studied further on. According to the basic setting-rules the parameters for the tripping characteristics at each line type are (Table 2):

| No. | $s$ | $I_p$ [kA] | $I_{p,ind}$[kA] | $T_p$ [s] | $a$ | $b$ [s] |
|---|---|---|---|---|---|---|
| 1. | 1.1 | 0.638 | 0.638 | 0.43 | -3.61 | 5.0 |
| 2. | 1.1 | 0.638 | 1.375 | 0.43 | -3.61 | 5.0 |
| 3. | 1.1 | 0.638 | 1.375 | 0.43 | -3.61 | 5.0 |
| 4. | 1.1 | 0.638 | 0.972 | 0.43 | -3.61 | 5.0 |

Table 2: 51 setting parameters for the IEC Very-Inverse and the log. characteristic for line type acc. Table 1

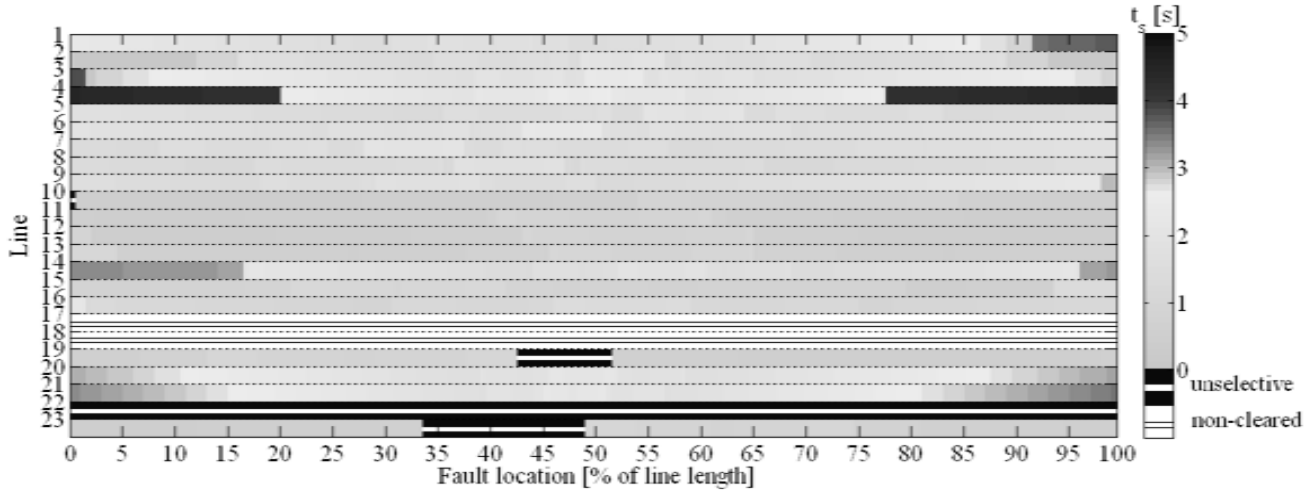



Figure 5: 51 fault clearing with IEC Very-Inverse characteristic for phase-to-phase faults

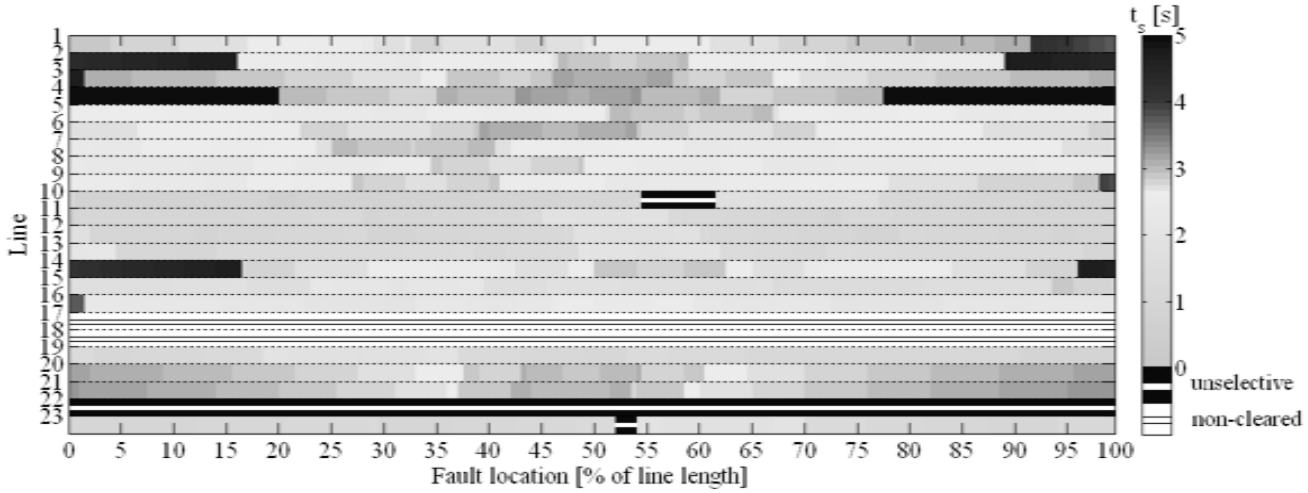



Figure 6: 51 fault clearing with logarithmic characteristic for phase-to-phase faults

Thereby the range of inverse operation is between $I_p$ and $20 \cdot I_p$ for both characteristics and $t_{p,min}$ is assumed to be 0.3s.
Figures 5 and 6 show the results. Selectivity can be ensured for almost all fault locations. For faults on lines 10, 19, 22 and 23 unselective tripping can be observed, i.e. more than the two main relays tripped during the clearing time interval.
However the logarithmic characteristic can also clear line 19 selective but on average the clearing time is higher. On lines 17 and 18 the main O/C relays failed to clear the faults.

### 5.3 Phase-to-earth faults

Line residual O/C relays are necessary as back-up protection for remote phase-to-earth faults and as protection against high resistance faults not detected by distance relays. They are typically directional (67N). The same characteristics as for phase-to-phase faults are studied.
Since a balanced operation causes no residual current the pickup current $I_p$ can be adjusted more sensitive and equal for all line types. $I_p$ is set to $0.2 \cdot I_{th,min}$. According to the basic setting-rules the other parameters for the tripping characteristics are (Table 3):

| No. | $I_p$[kA] | $I_{p,ind}$[kA] | $T_p$[s] | a | b[s] |
|---|---|---|---|---|---|
| 1.-4. | 0.120 | 0.120 | 1.69 | -2.0 | 5.0 |

Table 3: 67N setting parameters for the IEC Very-Inverse and the log. characteristic for line type acc. Table 1

Thereby the range of inverse operation is between $I_p$ and $20 \cdot I_p$ for the IEC characteristic and between $I_p$ and $80 \cdot I_p$ for the logarithmic characteristic. The minimum tripping time $t_{p,min}$ is assumed to be 1.2s.
Figures 7 and 8 show the results. One can see the IEC characteristic is less suitable for this concept. That is due to its narrow range of inverse operation and the wider spectrum of the residual short-circuit current. All O/C relays measuring a residual current above $20 \cdot I_p$ trip (unselective) at the same time. On average the tripping time is higher with the logarithmic characteristic indeed but selectivity can be ensured for almost all fault cases.
In comparison with Figure 6 the same lines are in trouble and it is possible to summarise the maloperation of the protection coordination concept to few practical constraints as to be discussed next.

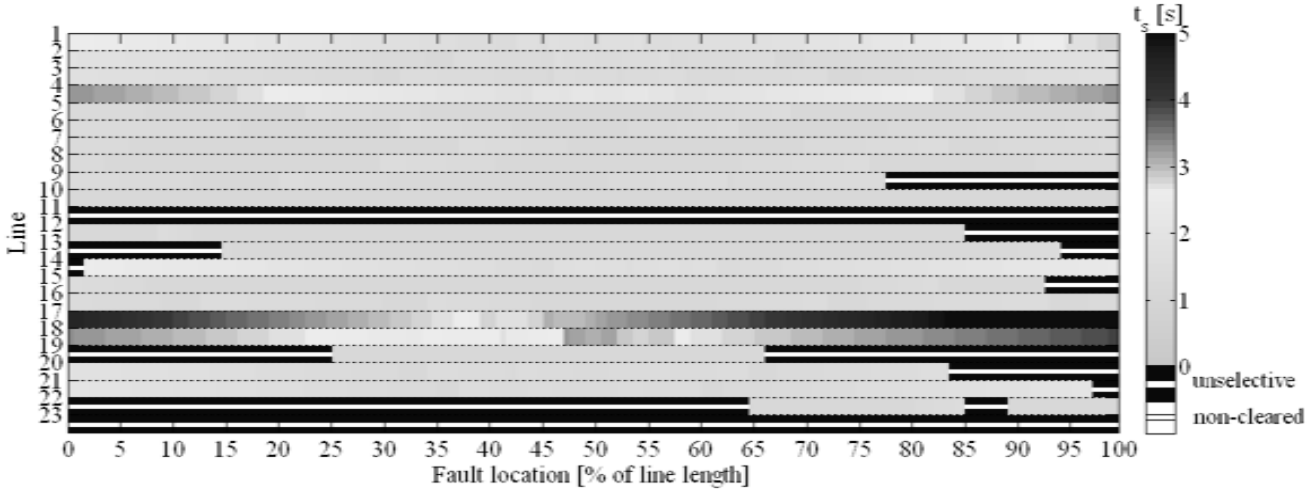



Figure 7: 67N fault clearing with IEC Very-Inverse characteristic for phase-to-earth faults

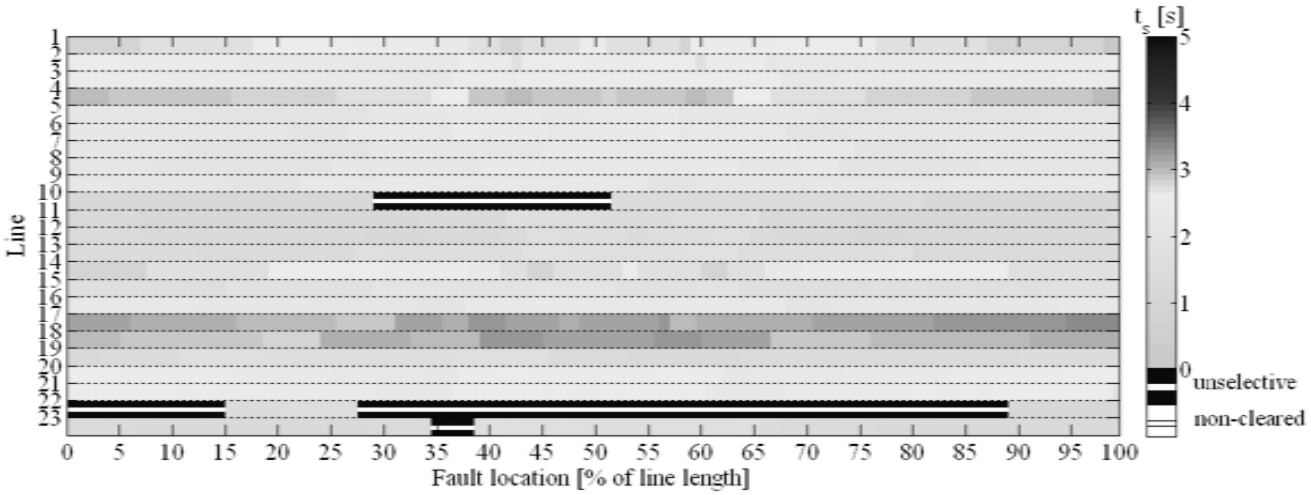



Figure 8: 67N fault clearing with logarithmic characteristic for phase-to-earth faults

## 6 Practical constraints of the proposed protection coordination concept

### 6.1 Passive radial feeder

The proposed protection coordination concept ensures selectivity if a required minimum current difference exists. That is lines without a local infeed at the remote bus can only be protected selectively if the remote bus has at least three feeders. Otherwise the seen short-circuit current is equal to all relevant O/C relays (Figure 9). That is the constraint for line 22 for phase-to-phase faults.

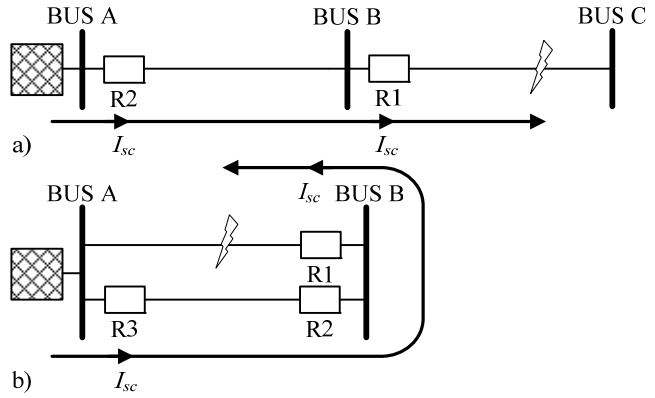



Figure 9: Passive radial feeder

### 6.2 Long lines with weak infeed at the remote bus

However a local infeed at the remote bus does not ensure successful protection coordination. Its contributed short-circuit current must cause the required minimum current difference. That is rather achieved the higher the line impedance and the stronger the infeed. But for very long lines with weak infeed at the remote bus the coordination concept can fail dependent on the fault location if the seen short-circuit current gets below pickup. That is the problem with lines 17 and 18 for phase-to-phase faults.

### 6.3 Short double-circuit lines with strong infeed at the remote bus

A meshed topology must be examined in detail if a bus with high short-circuit power (BUS B, Figure 10) is connected via a short double-circuit line with adjacent buses (BUS A).

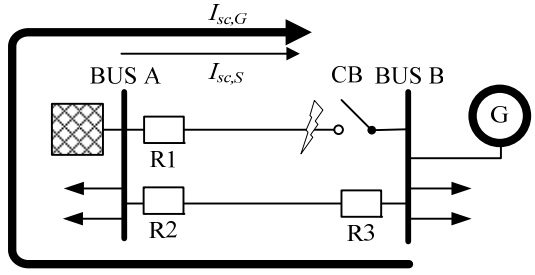



Figure 10: Short double-circuit line with strong infeed at the remote bus

Although the required current difference should be sufficient, the strong infeed at the remote BUS B defines the seen short-circuit current of the relays R1-R2 or R1-R3 to be coordinated respectively after operation of the circuit-breaker CB. Now the short-circuit current contribution $I_{sc,S}$ of the remaining network at BUS A is negligible little and the required current difference is not sufficient to ensure selectivity.

### 6.4 Timer sequence

The previous statements consider the network topology. The dynamic protection simulation with PSS™SINCAL made it possible to detect a practical constraint caused by the timer sequence of the O/C protection devices.
Timer are reset by the dropout value. Assuming that a circuit-breaker tripped and the short-circuit current escalates due to the changed topology all timer run uninterrupted. At this time the tripping times $t_p$ of the O/C relays decrease however the protection coordination keeps nominal selective. But dependent on the actual timer state $T$ these relays trip instantaneous for which the adjusted tripping time $t_p$ is below the timer state $T$. These relays trip unselectively.

## 7 Conclusion

It could be shown that software assisted protection coordination considers decisive benefits. Different concepts can be checked using only one simulation setup. An O/C protection coordination concept based on equal characteristics for all O/C relays was proposed and evaluated. The standard O/C characteristics according ANSI and IEC provide worse performance regarding selectivity but shorter fault clearing times with respect to the logarithmic characteristic. However its practical application depends on available device-specific features.
The evaluation gives rather reliable results based on real transmission network data. An appropriate visualisation method has been developed to illustrate the enormous amount of output data clearly arranged for further investigations. Based on these visualisation diagrams, practical constraints of protection coordination concepts can be easily detected and systematic solutions can be proposed. In this way a protection coordination concept can be evaluated properly and the procedure and the existing constraints can be documented understandably.

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